

NITTE (Deemed to be University)

(Established under Sec.(3) of UGC Act, 1956)

First Semester M.Tech (CBCS) Degree Examinations, December 2025

25MCT122 – MECHATRONICS SYSTEM DESIGN

(For MCT)

Duration: 3 Hours]

[Max. Marks: 100

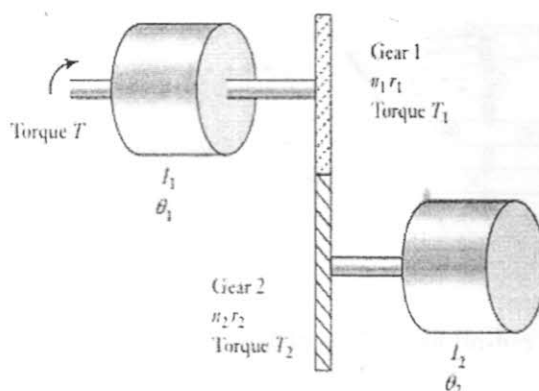
Note:

Answer **Five full** questions choosing **Two full** questions from **Module – I & Module – II each** and **One full** question from **Module – III**.

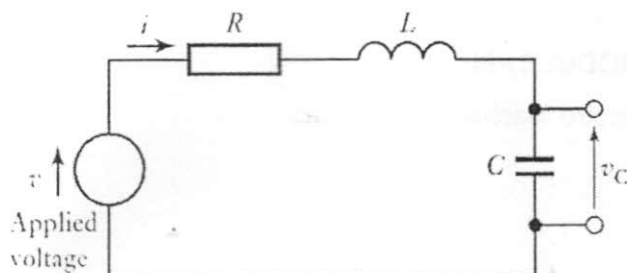
MODULE – I

Marks BT* CO* PO*

1. a) Derive the differential equation for the two-gear train system shown in figure 1(a) assuming negligible Moment of Inertia and damping coefficient



- b) Explain the working of LVDT and derive the voltage output equation of LVDT
2. a) Obtain the equation for pneumatic resistance, pneumatic capacitance and pneumatic inductance
- b) Explain the working of following sensor (i) Eddy current (ii) Capacitive (iii) Hall effect sensor
3. a) Derive the relationship between the potential difference across the resistor R and the input 'V' for the LCR circuit shown in the Fig.



10 3 1 1,2

10 2 2 1,2

10 2 1 1,2

10 2 2 1

10 3 1 1,2

- b) Derive the equation relating the strain with the gauge factor of the strain gauge, initial resistance of the strain gauge, input voltage and output voltage of the Wheatstone bridge circuit.

10

2

2

1,2

MODULE - II

4. a) Design a basic pneumatic Circuit for the Quarry stone sieving application as shown in Fig. Q4(a). Quarry stones are fed from a crushing roller to two vibrating sieves by means of an overhead conveyor belt. The fine upper sieves (1A) oscillate in opposing push-pull motion to the coarser lower screen (2A). Reversal takes place in the retracted end positions via two roller lever valves. A third single-acting cylinder (3A) unclogs the sieves via two cables. The stone sorter is switched on and off by a valve with selector switch

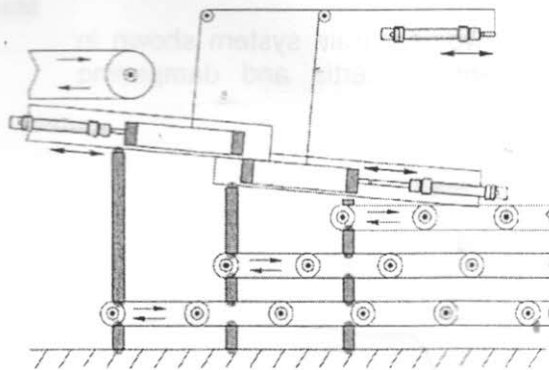


Fig 4(a)

- b) Design an active low-pass filter with a cut-off frequency of 100Hz.

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3

3

1,2

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3

4

1,2

5. a) Parts are to be stamped with a stamping device. By pressing two pushbutton switches the die is pushed down and the part is stamped. When the stamping pressure has been achieved, the die is returned to its start position. Design a electro-pneumatic system to implement this operation.
- b) A 4 bit R-2R ladder digital to analog converter has an input reference voltage of 8V. Derive the equation for the output of the R-2R ladder DAC. Obtain the analog output for all the possible combinations of 4 bit digital number.

10

3

3

1,2

10

3

4

1,2

6. a) Sketch and explain the H-Circuit to control the speed and direction of a DC Motor
- b) Derive the voltage out equation of summing amplifier and differential amplifier

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2

3

1

10

3

4

1,2

MODULE - III

7. a) Explain the benefits of Micro Electro Mechanical Systems.

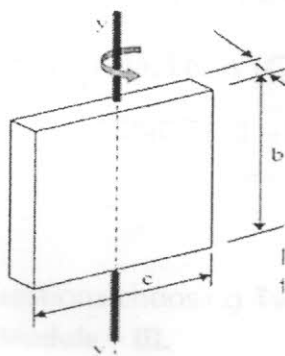
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2

5

1

- b) What would happen to the required torque to turn a micromirror as shown in Fig. Q7(b) with a 50% reduction in linear dimensions



OR

10 2 5 1

8. a) Explain the photolithography process.
b) Explain three different types of Chemical vapour deposition process used in MEMS fabrication.

10 2 5 1

10 2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
